# COMPLICATION SIGNAL SURVEILLANCE TEMPLATE

CSST — v1.0

| Site Name / Code | Protocol Number | Review Period (from — to) | Date of Review |
|---|---|---|---|
| | | | |

| Reviewer Name | Role / Title | Review Type |
|---|---|---|
| | | |

## A Case Volume Summary

| Parameter | Count |
|---|---|
| Total investigational procedures performed this period | |
| Total cumulative procedures (inception to date) | |
| Surgeons active this period | |
| New surgeons with first case in this period | |

## B Aggregate Complication Counts by Category

| Category | This Period (n) | Rate (%) | Cumulative (n) | Cumul. Rate (%) |
|---|---|---|---|---|
| Intraoperative — device-related (glenosphere, baseplate, humeral stem) | | | | |
| Intraoperative — surgical technique (positioning, exposure, preparation) | | | | |
| Intraoperative — unplanned implant configuration change | | | | |
| Early postoperative (≤90 days) — device-related (dislocation, notching, fracture) | | | | |
| Early postoperative (≤90 days) — wound or infection | | | | |
| Late postoperative (>90 days) — device-related | | | | |
| Device deficiency (device not functioning as intended) | | | | |
| **TOTAL — all categories** | | | | |

*Rate = n events / total procedures this period × 100. Source: all SCML entries for the review period indicated above.*

## C Signal Detection — Threshold Assessment

| Signal Test | Threshold | This Period Value | FLAG |
|---|---|---|---|
| Any single device-related category rate exceeds defined threshold | Suggested: 5% | | YES / NO |
| Total intraoperative complication rate exceeds defined threshold | Suggested: 10% | | YES / NO |
| Early dislocation rate (rTSA) exceeds defined threshold | Suggested: 5% | | YES / NO |

| | | | |
|---|---|---|---|
| Any category shows a rate at least double that of the preceding review period | Comparison-based | | YES / NO |
| New complication category with at least 2 events not seen in prior periods | Emergence-based | | YES / NO |
| Any individual surgeon with at least 2 events in a single category this period | Concentration-based | | YES / NO |

*Complete Section D for each cell flagged YES.*

## D Signal Assessment (complete only if any flag in Section C)

| Signal flagged | Events underlying | Clinical pattern | Surgeon(s) | Device lot(s) | Likely cause* | Action planned |
|---|---|---|---|---|---|---|
| | | | | | | |
| | | | | | | |
| | | | | | | |

**Cause categories: LC = Learning curve | TC = Technique deviation | DQ = Device quality concern | PI = Patient-related | UN = Undetermined*

## E Comparison with Previous Period

| Metric | Previous Period | This Period | Trend | Comment |
|---|---|---|---|---|
| Total complication rate | | | ↑ / ↓ / = | |
| Device-related complication rate | | | ↑ / ↓ / = | |
| Dislocation rate (rTSA) | | | ↑ / ↓ / = | |
| Infection rate | | | ↑ / ↓ / = | |

## F Actions and Escalation

- ■ No signal detected. No action required. Review documented for quality record.
- ■ Signal detected but assessed as expected or explainable. Rationale documented below.
- ■ Signal detected. Action initiated. See action table below.

Rationale (if applicable): _____

| Action | Responsible Party | Target Date | Done |
|---|---|---|---|
| Notify sponsor or CRO monitor | | | ■ |
| Escalate to SLCT intensified review for affected surgeon(s) | | | ■ |
| Convene site medical review meeting | | | ■ |
| Initiate device deficiency report to sponsor (ISO 14155:2020, Cl. 10.8(b)) | | | ■ |
| Other: | | | ■ |

## G Reviewer Certification

I certify that this aggregate review was conducted using all SCML entries for the review period indicated above, that the counts and rates are accurate to the best of my knowledge, and that flagged signals have been assessed and actioned as documented in this template.

| Primary Reviewer | Signature | Date |
|---|---|---|
| Co-Reviewer (if applicable) | Signature | Date |